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## Learning Check - Advanced Search Algorithms

≡ Instructions:

### Review Learning Objectives

1. Which of the following is a reason to use depth-first search (DFS) over breadth-first search (BFS)?

1. Which of the following is a reason to use depth-first search (DFS) over breadth-first search (BFS)?
- ☐ Depth-first search can find a solution that breadth-first search would not be able to.
  - ☐ Depth-first search can be used to solve a larger set of problems before starting the assignment? I'm here to help.
  - ☐ Depth-first search has a lower time complexity than BFS in the worst case.
  - ☒ Depth-first search has a lower space complexity than breadth-first search in the worst case.

Correct

Assignment details

DFS has space complexity  $O(bm)$ , while BFS has space complexity  $O(b^d)$ , where  $b$  is the maximum branching factor of the search tree,  $d$  is the depth of the least cost solution, and  $m$  is the maximum depth of the state space.

Due Date: Oct 4, 11:59 PM IST Attempts: 4 left (5 attempts every 24 hours)

Submitted  
Sep 29, 1:56 AM IST

Try again

2. Which of the following are reasons to use breadth-first search (BFS) over depth-first search (DFS)? Select all that apply.

1 / 1 point

**Your grade**

☐ Breadth-first search has to explore up to  $2^{100}$  nodes in the worst case.

☒ BFS is complete even for **100%** depth search space.

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see feedback

Correct  
DFS may get stuck in an infinite depth graph while BFS always expands the shallowest node and thus it is complete.

☒ BFS is cost optimal for problems with unit costs (all actions have the same cost).

☒ **Correct**  
BFS will expand from the shortest path in the graph with uniform cost.

☐ BFS always finds the solution faster than DFS when the search space is small.

3. If you are working with restricted memory, it would be preferable to use

1 / 1 point

- ☐ Breadth-first search
- ☒ Depth-first search
- ☒ **Correct**
- The space complexity of BFS is  $O(b^l)$ , while DFS has a lower complexity of  $O(bm)$ . BFS requires tracking of all node layers while DFS only stores one path's node at a time.